Introduction to the Theory of Computation

Lecture 14. Post Correspondence Problem

Neo (Yuan) Li

Oct 2026

Fudan University

yuan_li@fudan.edu.cn

Are There "Natural" Undecidable Problems?

- So far, every undecidable language we've seen (L_{halt}, L_{flip}, L_{accept}, ...) is **about Turing machines**: the inputs are TM descriptions, and membership depends on how the TM behaves.
- **Question:** are there undecidable problems that are **natural** — stated purely in terms of combinatorics, algebra, number theory, ..., with no mention of machines or programs?

Some “Natural” Undecidable Problems

1. Post Correspondence Problem
2. Hilbert’s 10th Problem
3. Word Problem for Groups
4. Wang Tiling Problem
5. Ambiguity / Equivalence / Universality of CFGs
6. Matrix Mortality Problem

   ⋮

Word Problem for Groups

- A **finitely presented group**:

$$G = \langle g_1, \ldots, g_n \mid r_1, \ldots, r_m \rangle$$

  finitely many **generators** g_i, finitely many **relations** $r_j = 1$.
- **Word Problem:** given G and a word w over $g_i^{\pm 1}$, is $w = 1$ in G?
- **Undecidable** (Novikov 1955, Boone 1958) — even for a single fixed G.

Word Problem: Examples

Let $G = \langle a, b \mid aba^{-1}b^{-1} \rangle$, i.e., $ab = ba$.

- **Example 1:** $w = a\,b\,a^{-1}\,b^{-1}$

$$a\,b\,a^{-1}b^{-1} \;\to\; b\,a\,a^{-1}b^{-1} \;\to\; b\,b^{-1} \;\to\; 1$$

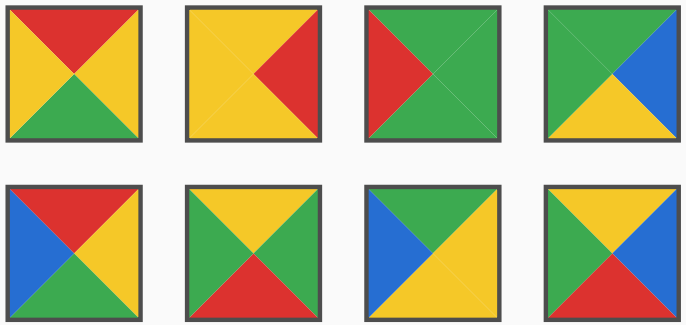
  So $w = 1$. ✓

- **Example 2:** $w = a\,b\,a^{-1}\,b$
  Invariant: (# of b) − (# of b^{-1}) never changes under the rules.
  It is 2 for w but 0 for 1, so $w \neq 1$. ×

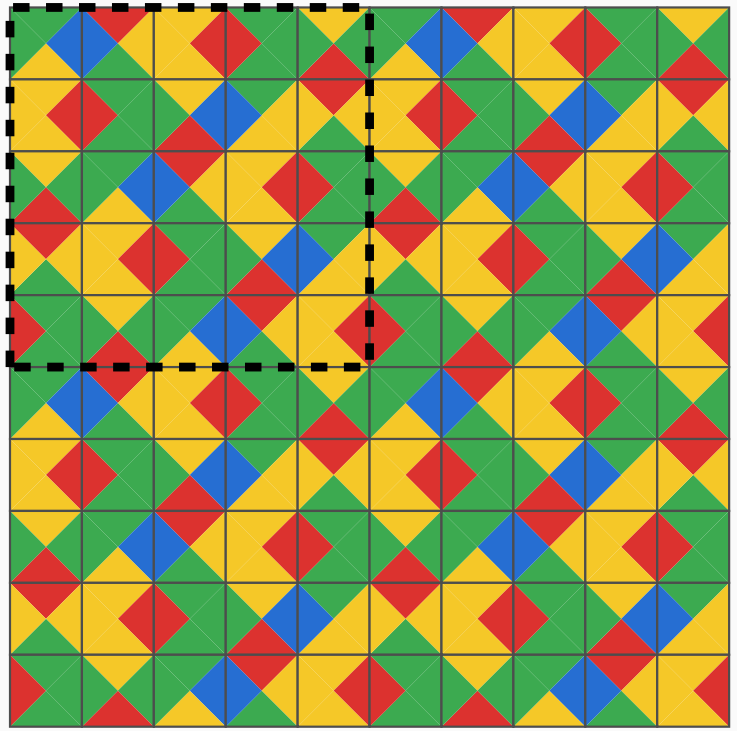
Wang Tiling Problem

- A **Wang tile**: a unit square with a color on each edge. No rotation or reflection.
- A **tiling**: put a tile at every point of $\mathbb{Z}^2$ so that **adjacent edges have matching colors**.
- **Problem:** given a finite set T of tiles (unlimited copies of each), can T tile the whole plane?
- **Undecidable** (Berger 1966).

Wang Tiling Problem: An Example



A set of 8 Wang tiles



A valid tiling: the dashed 5×5 block repeats in both directions forever

Adjacent tiles must match colors on shared edges.

Wang Tiling Problem: History

- **Hao Wang** (1961): conjectured that every tile set that tiles the plane can tile it **periodically**.
- True $\Rightarrow$ problem **decidable**.
- **Robert Berger** (1966), Wang's student: found an **aperiodic** tile set, disproving the conjecture, and proved the problem **undecidable**.

Hilbert's 10th Problem

- **Diophantine equation:** $p(x_1, \ldots, x_n) = 0$, where p is a polynomial with **integer coefficients**.
- **Hilbert (1900):** find a procedure that decides, for any such equation, whether it has an **integer** solution.

Hilbert's 10th Problem: Examples

- $x^2 + y^2 = 5$: **has** an integer solution.
  - Take $(x, y) = (1, 2)$: $1^2 + 2^2 = 5$. ✓
- $x^2 + y^2 = 3$: **no** integer solution.
  - Every square is $\equiv 0$ or $1 \pmod 4$.
  - So $x^2 + y^2 \equiv 0, 1$, or $2 \pmod 4$, never 3.

Hilbert's 10th Problem: History

- **1900:** Hilbert asks for an algorithm, expecting a **positive** answer.
- **1936:** Turing and Church make "algorithm" precise.
- **1950s–60s:** Davis, Putnam, Robinson.
- **1970:** Matiyasevich completes the proof: **undecidable**.

MRDP Theorem: Hilbert's 10th Problem is **undecidable**.

Post Correspondence Problem

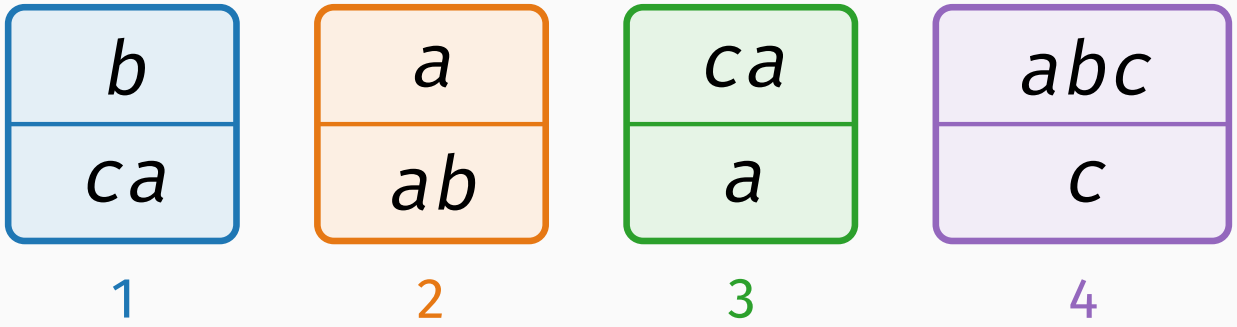
A **domino** $\left[\frac{t}{b}\right]$ is a pair of strings $t, b \in \Sigma^*$.

Given dominos $\left[\frac{t_1}{b_1}\right], \ldots, \left[\frac{t_k}{b_k}\right]$, a **match** is a sequence $s_1, \ldots, s_m \in \{1, \ldots, k\}$, $m \geq 1$ (repetition allowed), such that

$$t_{s_1} t_{s_2} \cdots t_{s_m} = b_{s_1} b_{s_2} \cdots b_{s_m}.$$

Post Correspondence Problem: given a set of dominos, does a match exist?

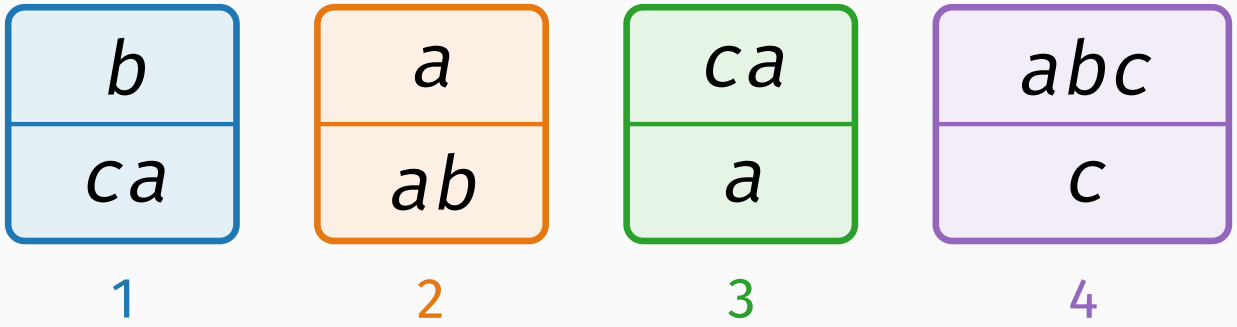
PCP: An Example with a Match

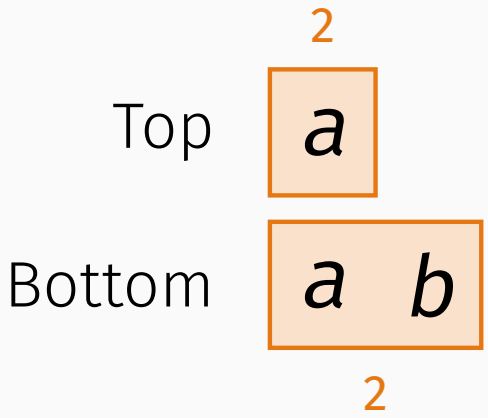


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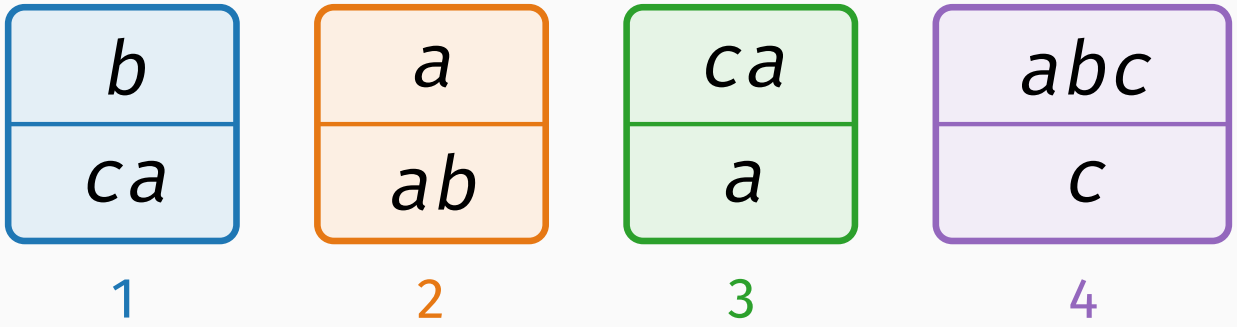
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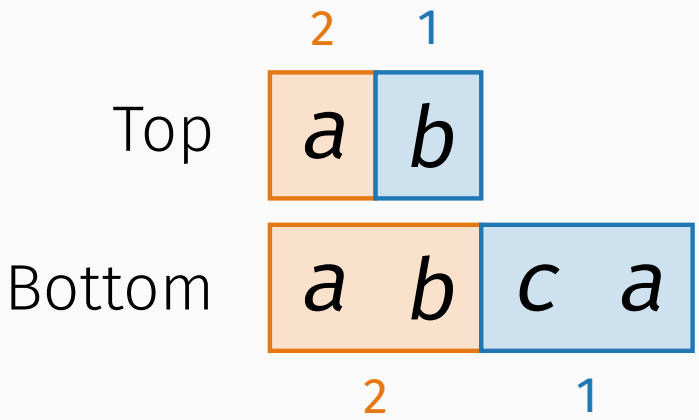
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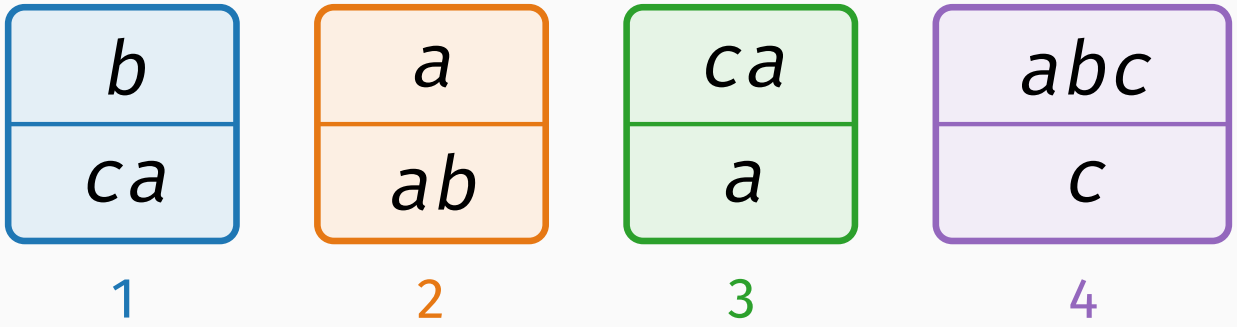


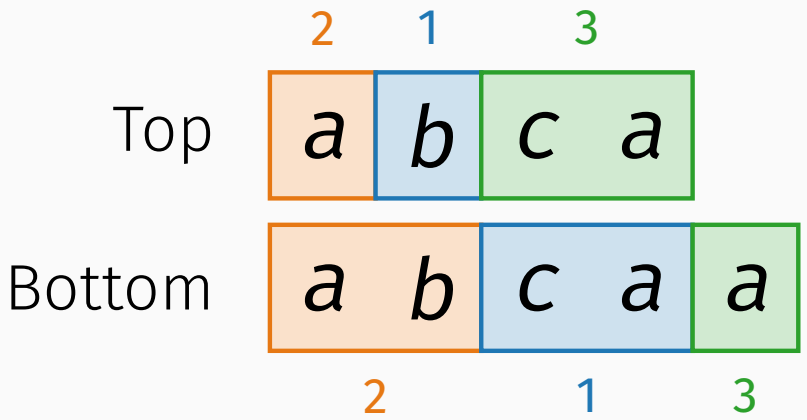
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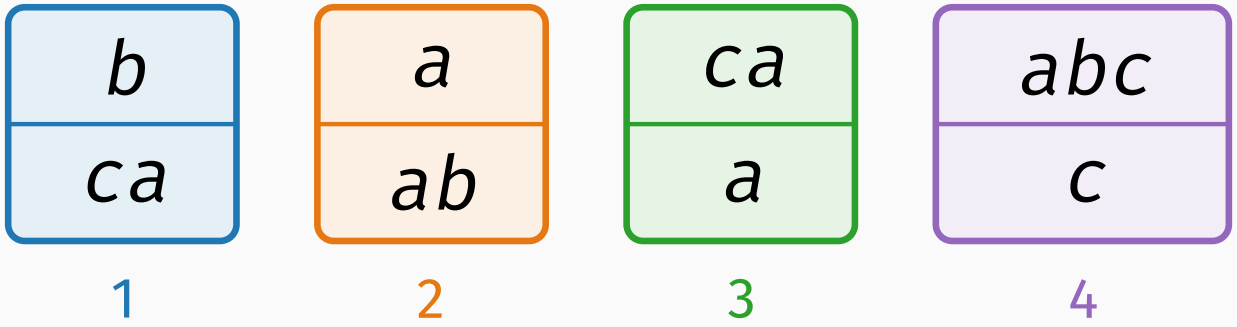


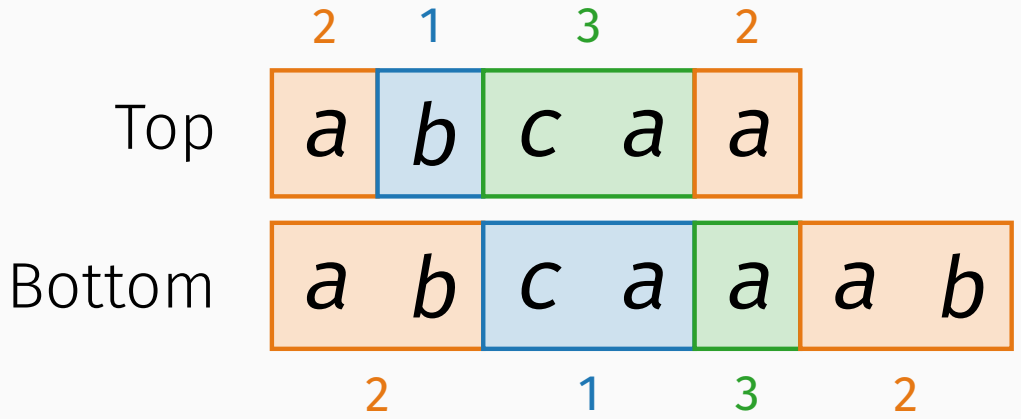
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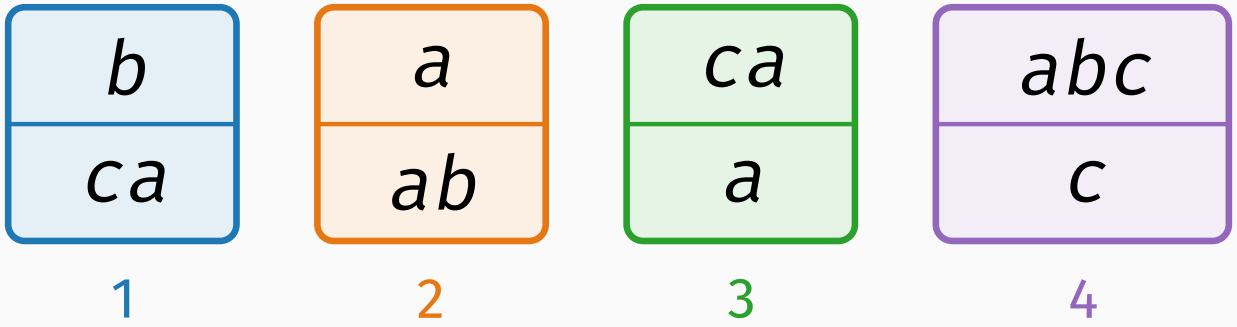


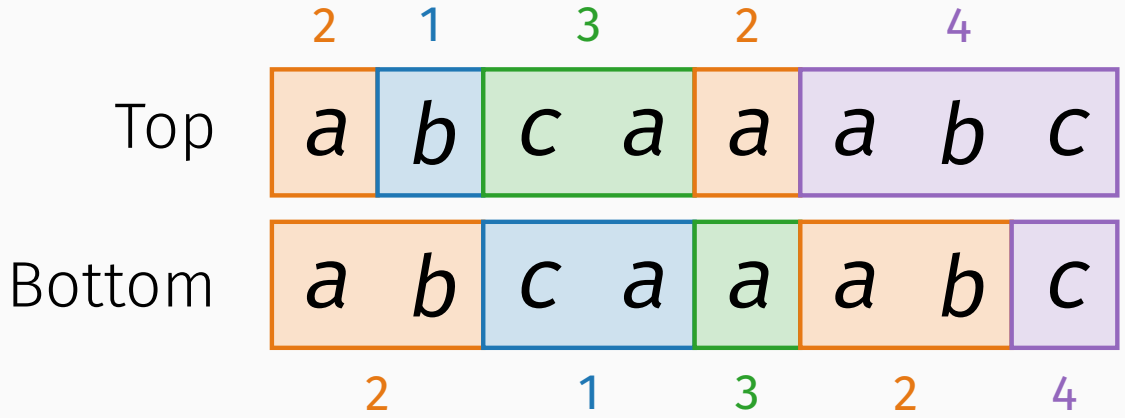
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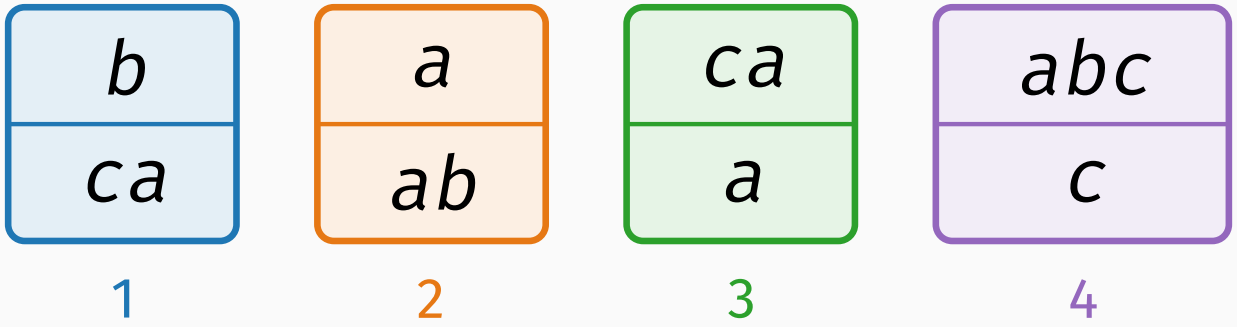


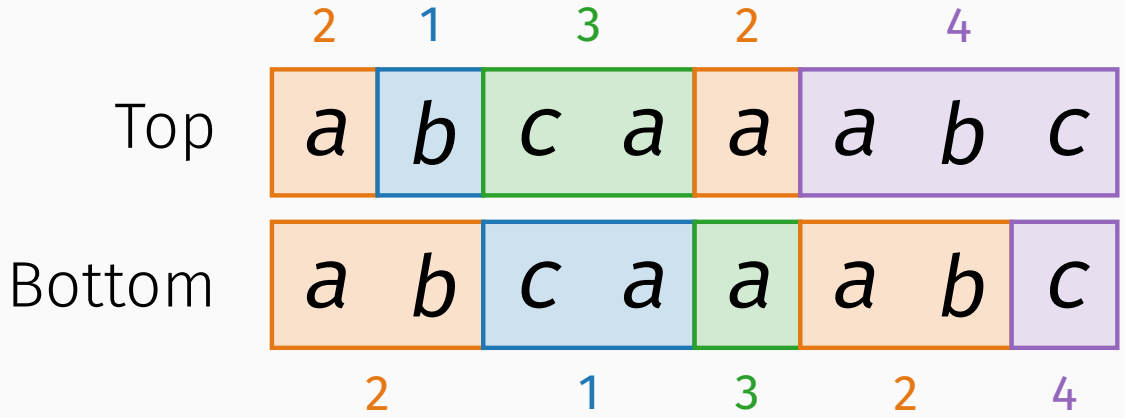
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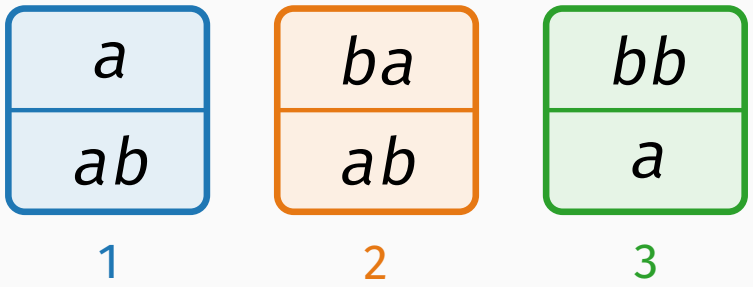
PCP: An Example with a Match





Top = Bottom = *abcaaabc*: a match!

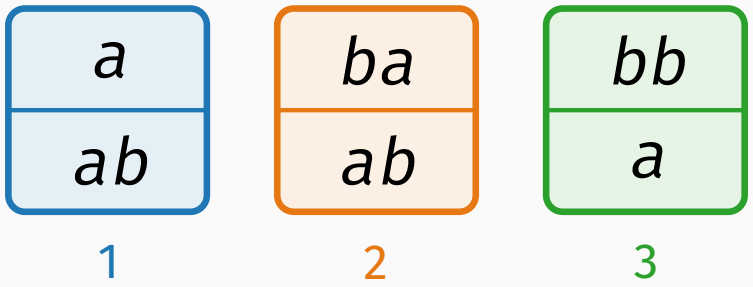
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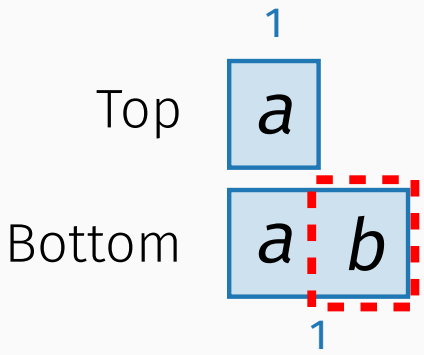


Top

Bottom

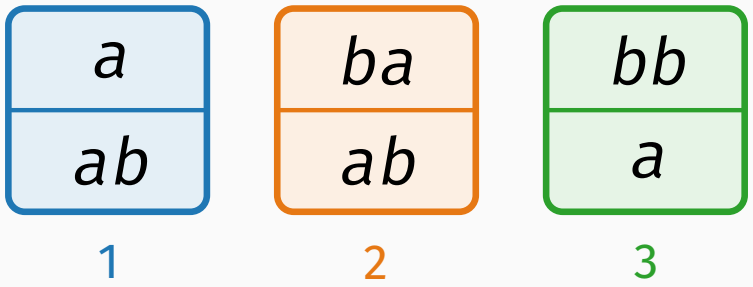
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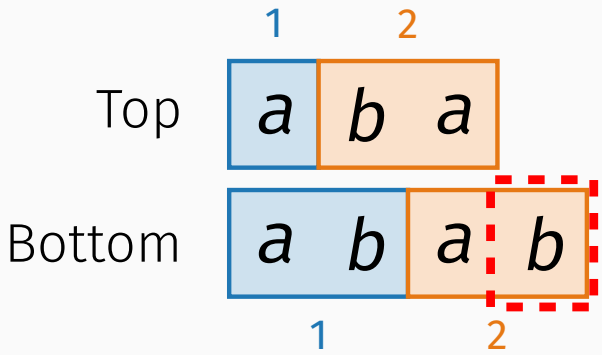




- The first domino must be 1: dominos 2 and 3 start with different letters on top and bottom.

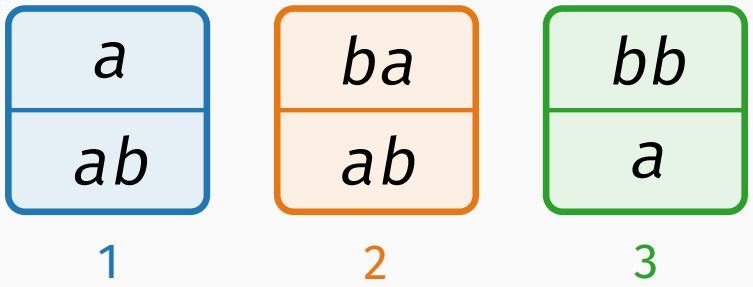
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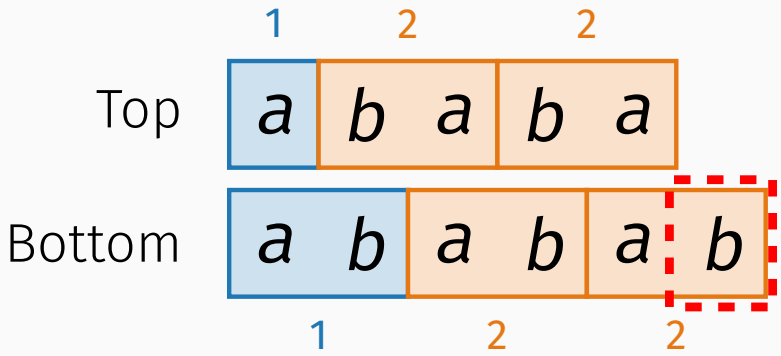




- The first domino must be 1: dominos 2 and 3 start with different letters on top and bottom.
- If the bottom is ahead by exactly *b*, the next domino must be 2 (domino 3 gives *bb* vs. *ba*), and afterwards the bottom is **again** ahead by exactly *b*.

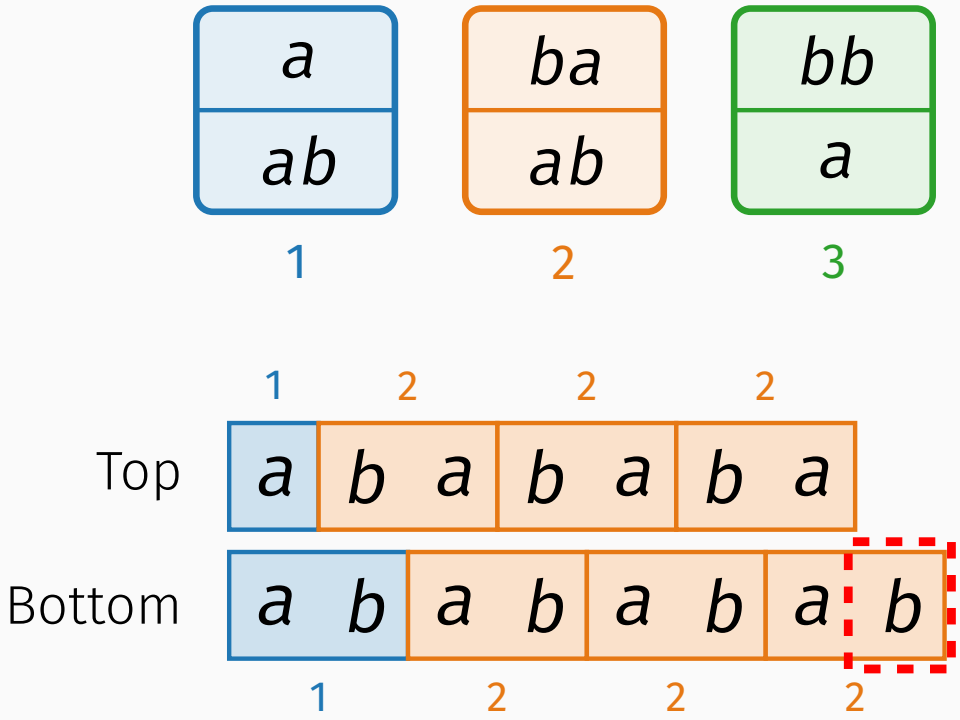
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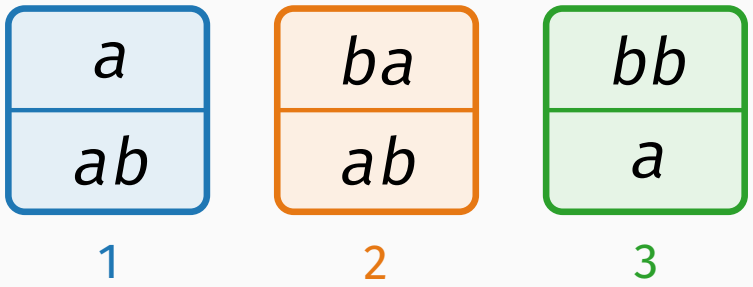
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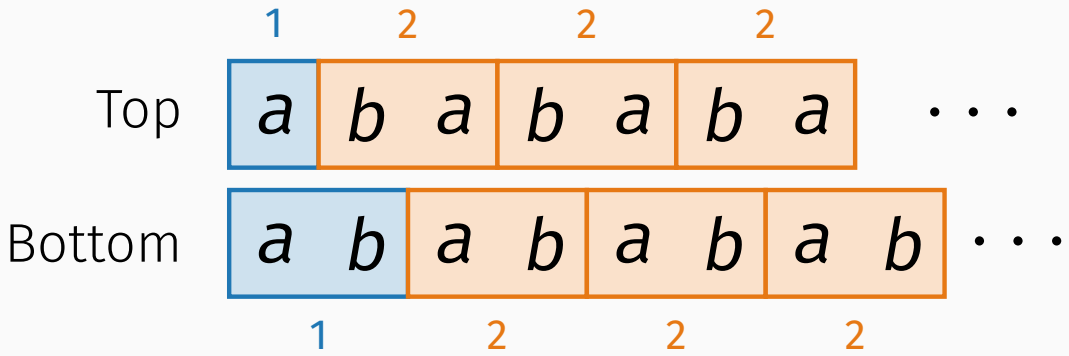
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PCP: An Example with No Match





- The first domino must be 1: dominos 2 and 3 start with different letters on top and bottom.
- If the bottom is ahead by exactly *b*, the next domino must be 2 (domino 3 gives *bb* vs. *ba*), and afterwards the bottom is **again** ahead by exactly *b*.
- So the bottom is **always one letter ahead**: no match exists.

Exercises

Does each collection have a match? If yes, find one.

1. $\left\{ \left[\frac{ab}{a}\right], \left[\frac{b}{bb}\right] \right\}$
2. $\left\{ \left[\frac{abc}{ab}\right], \left[\frac{ca}{a}\right], \left[\frac{acc}{ba}\right] \right\}$
3. $\left\{ \left[\frac{a}{baa}\right], \left[\frac{ab}{aa}\right], \left[\frac{bba}{bb}\right] \right\}$
4. $\left\{ \left[\frac{a}{ba}\right], \left[\frac{ba}{b}\right], \left[\frac{ba}{ab}\right] \right\}$
5. $\left\{ \left[\frac{1}{111}\right], \left[\frac{10111}{10}\right], \left[\frac{10}{0}\right] \right\}$

The Language L_{pcp}

Definition 1

Let $P = \left(\left[\frac{t_1}{b_1} \right], \ldots, \left[\frac{t_k}{b_k} \right] \right)$ be a collection of dominos.

$$L_{\text{pcp}} = \{ P : P \text{ has a match} \}.$$

Theorem 2 (Post, 1946)

L_{pcp} is undecidable.

Undecidability of PCP: Proof Idea

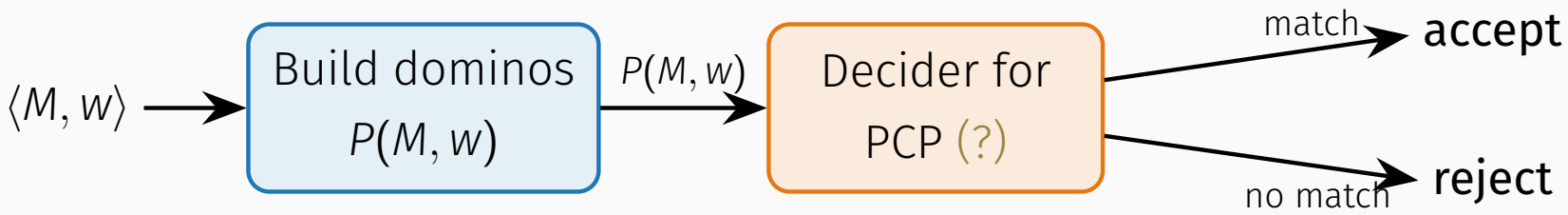
- **Reduce** L_{accept} to L_{pcp}: from M and w, build dominos $P(M, w)$ with

$$M \text{ accepts } w \iff P(M, w) \text{ has a match.}$$

- **Idea:** a match spells out an **accepting computation history** of M on w, one configuration at a time.

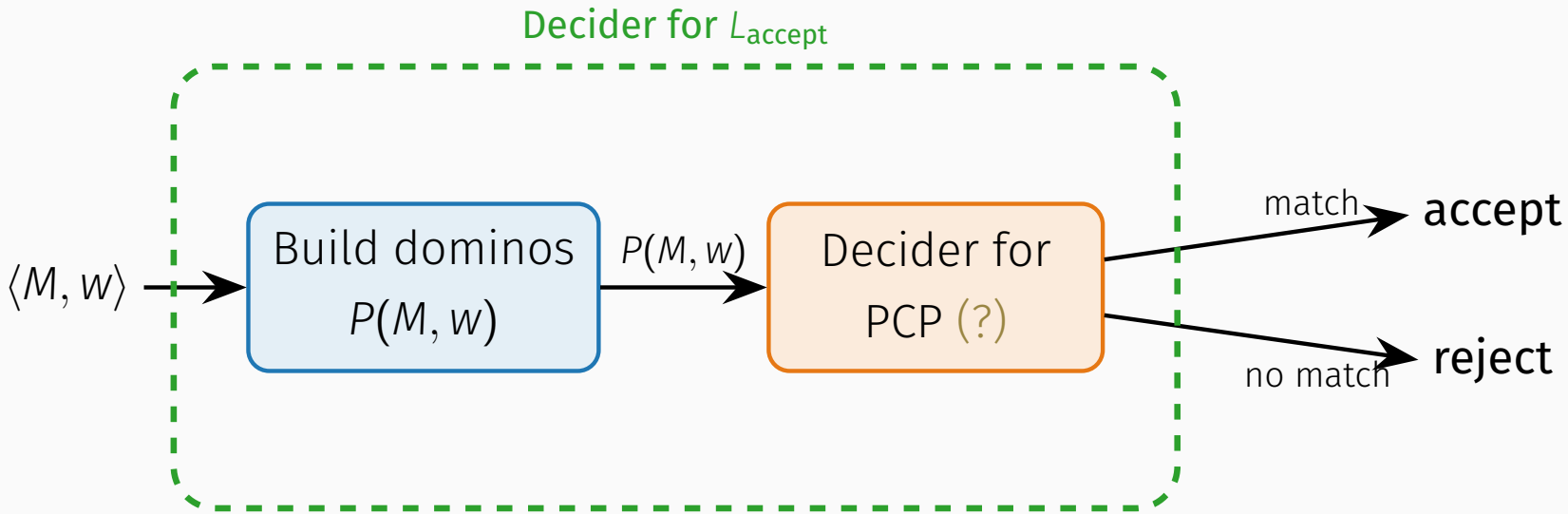
We follow Michael Sipser's proof (*Introduction to the Theory of Computation*, Sec. 5.2).

Undecidability of PCP: The Big Picture



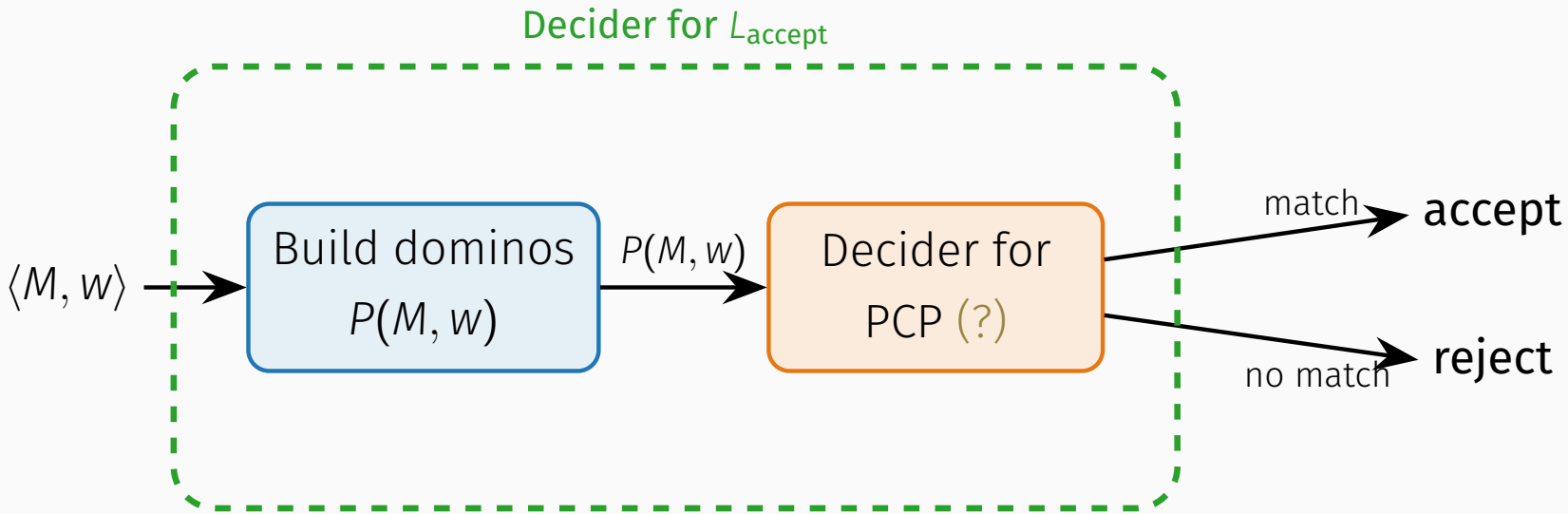
- Key property: M accepts $w \iff P(M, w)$ has a match.

Undecidability of PCP: The Big Picture



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- A decider for PCP would give a decider for L_{accept}.

Undecidability of PCP: The Big Picture



- Key property: M accepts $w \iff P(M, w)$ has a match.
- A decider for PCP would give a decider for L_{accept}.
- But L_{accept} is undecidable. So PCP is undecidable.

Setup and Assumptions

- M: single-tape TM, tape infinite to the right only.
- **Assumption 1:** M never moves its head off the **left end**.
- **Assumption 2:** if $w = \epsilon$, use $w = \sqcup$ instead.

Modified PCP (MPCP)

- **MPCP:** same as PCP, but the match must **start with the first domino** $\left[\frac{t_1}{b_1}\right]$.

$$L_{\text{mpcp}} = \{\, P : P \text{ has a match that starts with } \left[\tfrac{t_1}{b_1}\right] \,\}$$

- **Plan:**

$$L_{\text{accept}} \longrightarrow L_{\text{mpcp}} \longrightarrow L_{\text{pcp}}$$

From MPCP to PCP: the $\star$ Trick

Notation ($u = abc$):

$\star u = \star a \star b \star c,\ u\star = a \star b \star c\star,\ \star u\star = \star a \star b \star c \star .$

Given MPCP instance $P' = \left\{ \left[\frac{t_1}{b_1}\right], \ldots, \left[\frac{t_k}{b_k}\right] \right\}$, let

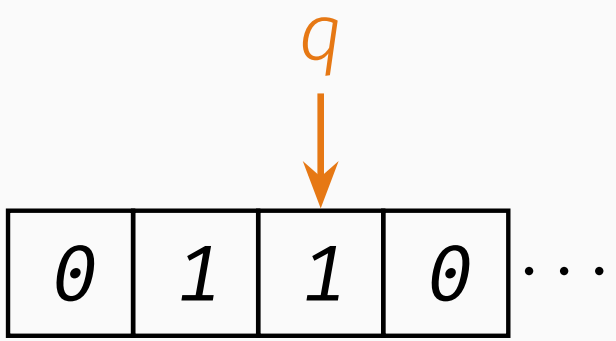
$$P = \left\{ \left[\frac{\star t_1}{\star b_1 \star}\right], \left[\frac{\star t_1}{b_1 \star}\right], \ldots, \left[\frac{\star t_k}{b_k \star}\right], \left[\frac{\star \Diamond}{\Diamond}\right] \right\}.$$

- Only $\left[\frac{\star t_1}{\star b_1 \star}\right]$ starts with $\star$ on both sides $\Rightarrow$ every match **starts with it**.
- $\left[\frac{\star \Diamond}{\Diamond}\right]$ absorbs the extra $\star$ at the end.

P' has a match starting with $\left[\frac{t_1}{b_1}\right] \iff P$ has a match.

Turing Machine Configurations

- $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}})$.
- A **configuration** is one string: write the state **just left of the scanned cell**.



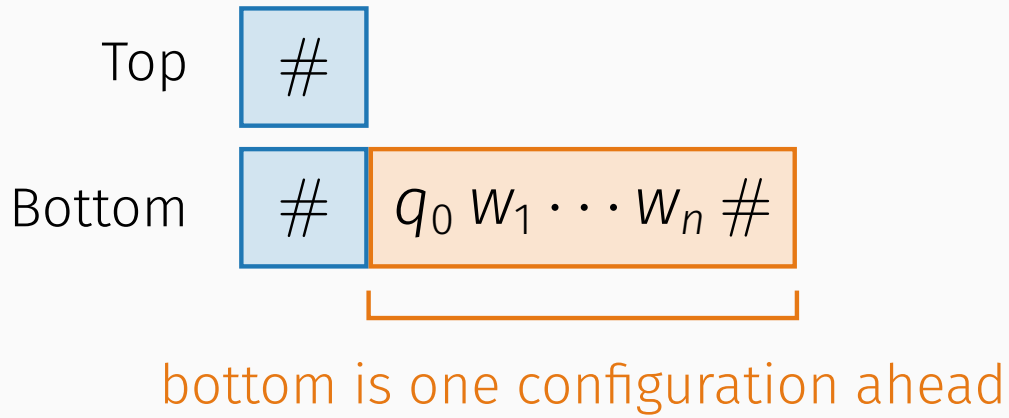
$\Longrightarrow \quad 01q10$

In general: $x_1 \cdots x_{i-1}\, q\, x_i \cdots x_n$.

Construction, Part 1: Initial Configuration

The **first domino** of P' writes the start configuration $q_0 w$ on the bottom ($w = w_1 \cdots w_n$):

$$\left[\frac{t_1}{b_1}\right] = \left[\frac{\#}{\# \, q_0 \, w_1 w_2 \cdots w_n \, \#}\right].$$



Construction, Part 2: Right Moves

For every $q \notin \{q_{\text{accept}}, q_{\text{reject}}\}$ and $a \in \Gamma$ with $\delta(q, a) = (r, b, R)$, put

$$\left[\begin{matrix} q & a \\ b & r \end{matrix}\right] \text{ into } P'.$$

Example: $\delta(q, 1) = (r, 0, R)$ gives $\left[\begin{matrix} q & 1 \\ 0 & r \end{matrix}\right]$:

$$0\ q\ 1\ 1 \quad \longrightarrow \quad 0\ 0\ r\ 1$$

Construction, Part 3: Left Moves

For every $q \notin \{q_{\text{accept}}, q_{\text{reject}}\}$ and $a \in \Gamma$ with $\delta(q, a) = (r, b, L)$, and **every** $c \in \Gamma$, put

$$\left[\frac{c\,q\,a}{r\,c\,b}\right] \text{ into } P'.$$

Example: $\delta(q, 1) = (r, 0, L)$, $c = 1$ gives $\left[\frac{1\,q\,1}{r\,1\,0}\right]$:

$$0\,1\,q\,1 \quad \longrightarrow \quad 0\,r\,1\,0$$

Construction, Part 4–5: Copying and Blanks

Part 4 (copy tape symbols): for every $a \in \Gamma$, put

$$\left[\frac{a}{a}\right] \text{ into } P'.$$

Part 5 (copy separator, or add a blank): put

$$\left[\frac{\#}{\#}\right] \quad \text{and} \quad \left[\frac{\#}{\sqcup\#}\right] \text{ into } P'.$$

Example: $0\,1\,q\,\# \longrightarrow 0\,1\,q\,\sqcup\,\#$

Construction, Part 6–7: Finishing an Accepting Run

Part 6 (let q_{accept} eat neighbors): for every $a \in \Gamma$, put

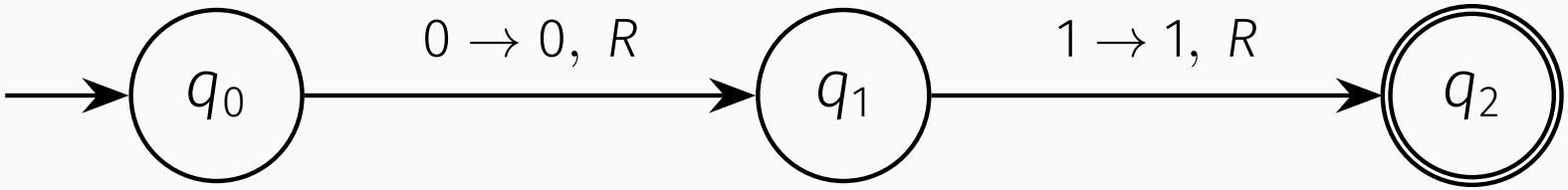
$$\left[\frac{a\, q_{\text{accept}}}{q_{\text{accept}}}\right] \quad \text{and} \quad \left[\frac{q_{\text{accept}}\, a}{q_{\text{accept}}}\right] \text{ into } P'.$$

Part 7 (close the match): put

$$\left[\frac{q_{\text{accept}}\, \# \, \#}{\#}\right] \text{ into } P'.$$

Example: $0\, q_{\text{accept}}\, 1 \to q_{\text{accept}}\, 1 \to q_{\text{accept}}$, then Part 7.

Example: A Tiny Turing Machine



Run on $w = 01$ $(q_2 = q_{\text{accept}})$:

$$\underbrace{q_0\,0\,1}_{C_0} \vdash \underbrace{0\,q_1\,1}_{C_1} \vdash \underbrace{0\,1\,q_2}_{C_2} \quad \text{accept}$$

Example: The Dominos of P'

Part 1 $\frac{\#}{\#q_0 01\#}$

Part 2 $\frac{q_0 0}{0 q_1}$ $\frac{q_1 1}{1 q_2}$

Part 4 $\frac{0}{0}$ $\frac{1}{1}$ $\frac{\sqcup}{\sqcup}$

Part 5 $\frac{\#}{\#}$ $\frac{\#}{\sqcup\#}$

Part 6 $\frac{0 q_2}{q_2}$ $\frac{1 q_2}{q_2}$ $\frac{\sqcup q_2}{q_2}$ $\frac{q_2 0}{q_2}$ $\frac{q_2 1}{q_2}$ $\frac{q_2 \sqcup}{q_2}$

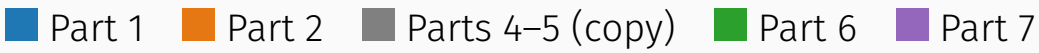
Part 7 $\frac{q_2 \# \#}{\#}$

Faded dominos exist in P' but are **not used** in the match below. (No left moves $\Rightarrow$ no Part 3.)

Example: Assembling the Match

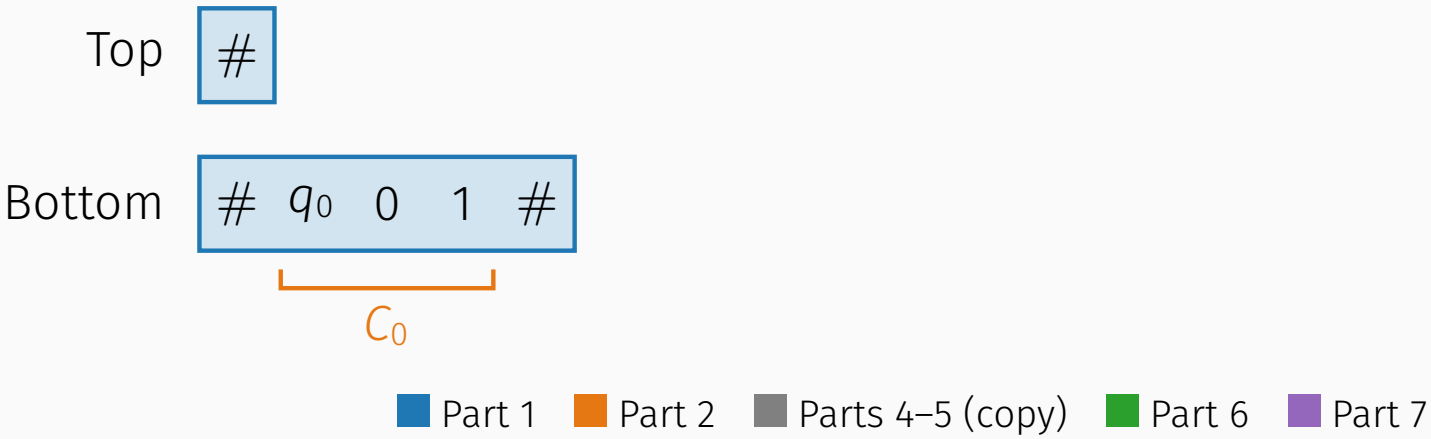
Top

Bottom



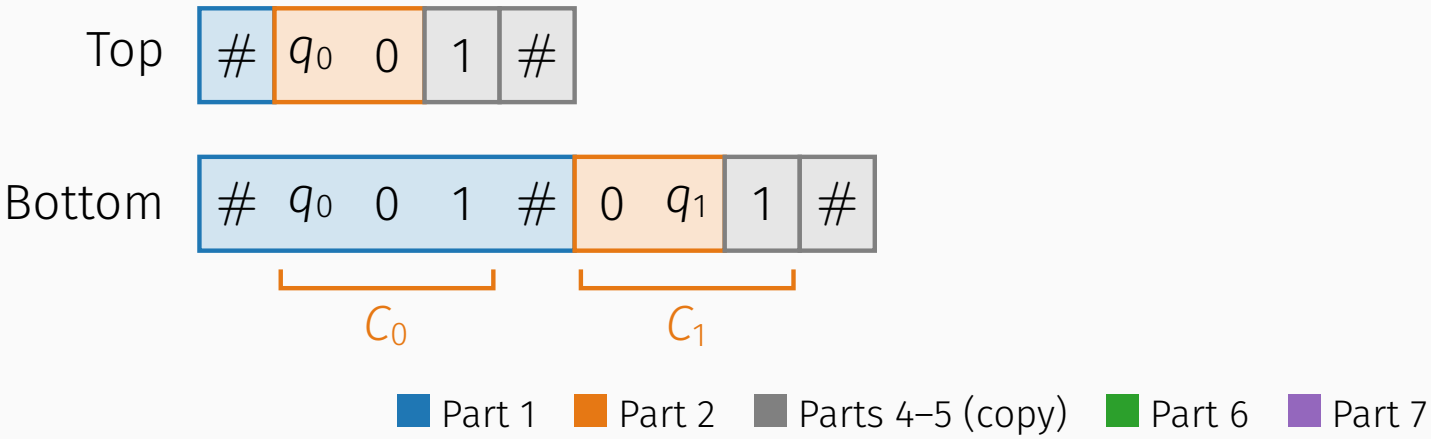
Start from the empty rows.

Example: Assembling the Match



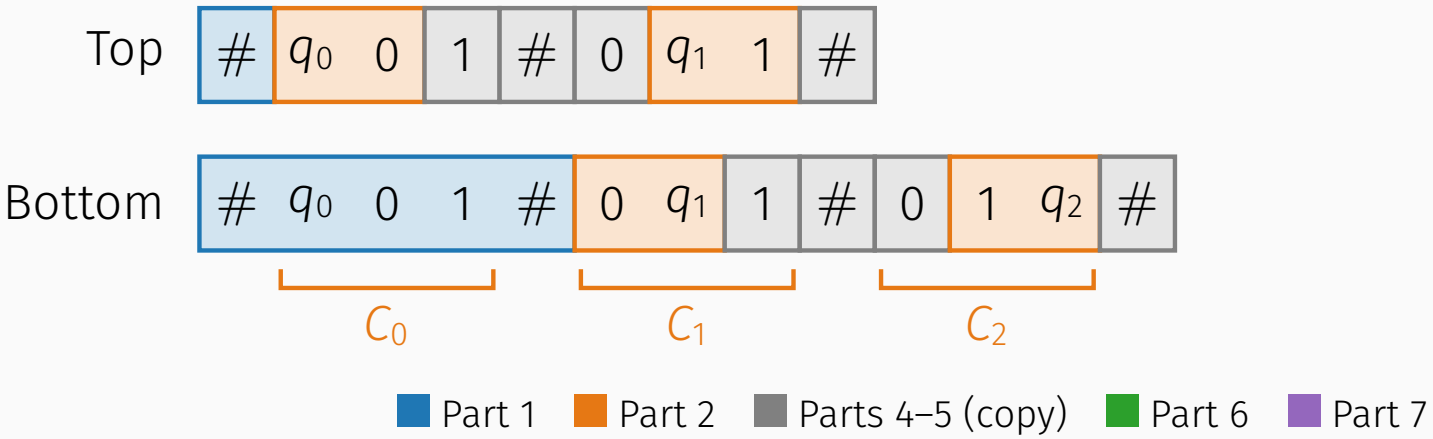
Part 1: bottom writes $C_0 = q_0\,0\,1$.

Example: Assembling the Match



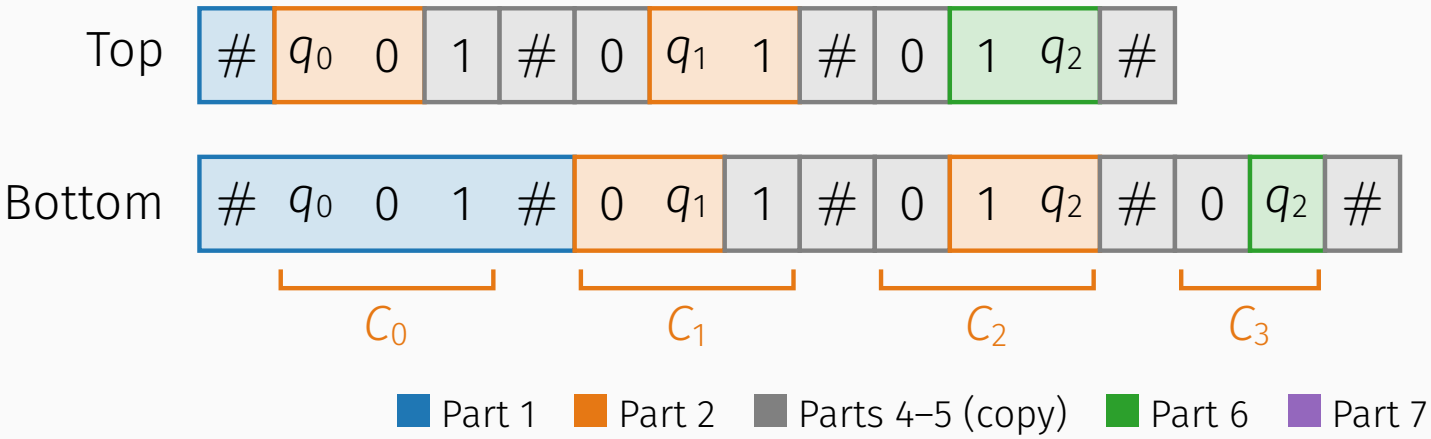
Top reads C_0; bottom writes $C_1 = 0\, q_1\, 1$.

Example: Assembling the Match



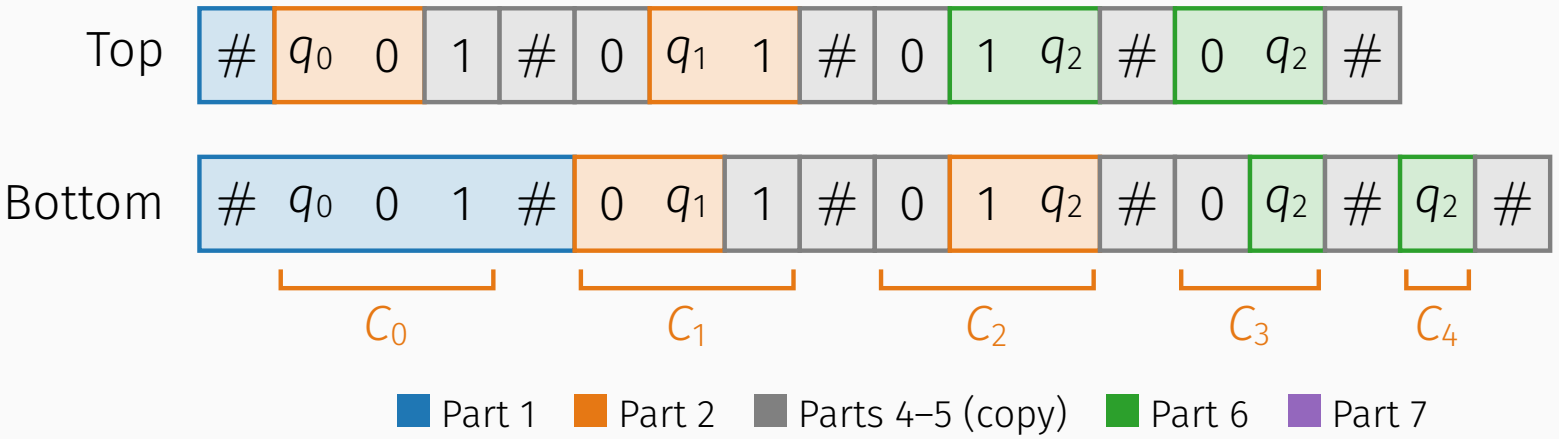
Top reads C_1; bottom writes $C_2 = 0\,1\,q_2$. **Accepted!**

Example: Assembling the Match



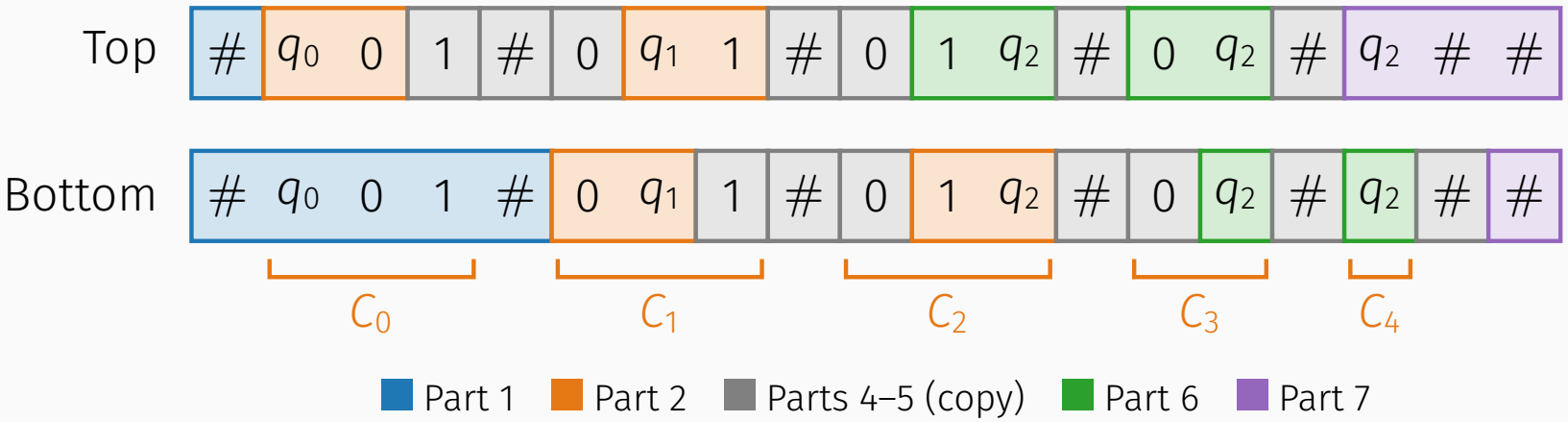
Part 6: q_2 eats 1; bottom writes $C_3 = 0\,q_2$.

Example: Assembling the Match



Part 6: q_2 eats 0; bottom writes $C_4 = q_2$.

Example: Assembling the Match



Part 7: top catches up. Top = Bottom: a match!

Example: The Matched String

Top $=$ Bottom $=$

$$\# \underbrace{q_0\,0\,1}_{C_0} \# \underbrace{0\,q_1\,1}_{C_1} \# \underbrace{0\,1\,q_2}_{C_2} \# \underbrace{0\,q_2}_{C_3} \# \underbrace{q_2}_{C_4} \#\#$$

- C_0, C_1, C_2: the accepting run of M on w.
- C_3, C_4: clean-up, so the top can catch up.

The match *is* the accepting computation history.

Correctness: If M Accepts w, Then P' Has a Match

Let $C_0 \vdash C_1 \vdash \cdots \vdash C_\ell$ be the accepting run of M on w.

- **Start:** d_1 writes $\#C_0\#$ on the bottom.
- **Round** i ($i < \ell$): top reads $C_i\#$, bottom writes $C_{i+1}\#$ (one Part 2/3 domino at the head, Part 4–5 elsewhere).
- **Clean-up:** Part 6 shrinks C_ℓ to q_{accept}; Part 7 closes.

$$\#C_0\#C_1\#\cdots\#C_\ell\#\cdots\#q_{\text{accept}}\#\#$$

Correctness: If P' Has a Match, Then M Accepts w

Fix a match $D_1 = d_1, D_2, \ldots, D_m$ of P', with common string W.

- Top_k, Bot_k: tops / bottoms of $D_1, \ldots, D_k$ concatenated. Both are **prefixes of** W.
- **Configuration string** $S \in \Gamma^* Q \Gamma^*$: exactly one state, no $\#$.
- $S \vdash S'$: one step of M (cells right of S hold $\sqcup$).
- $S \approx S'$: equal up to trailing $\sqcup$'s.
- The run of M on w: $C_0 = q_0 w \vdash C_1 \vdash C_2 \vdash \cdots$

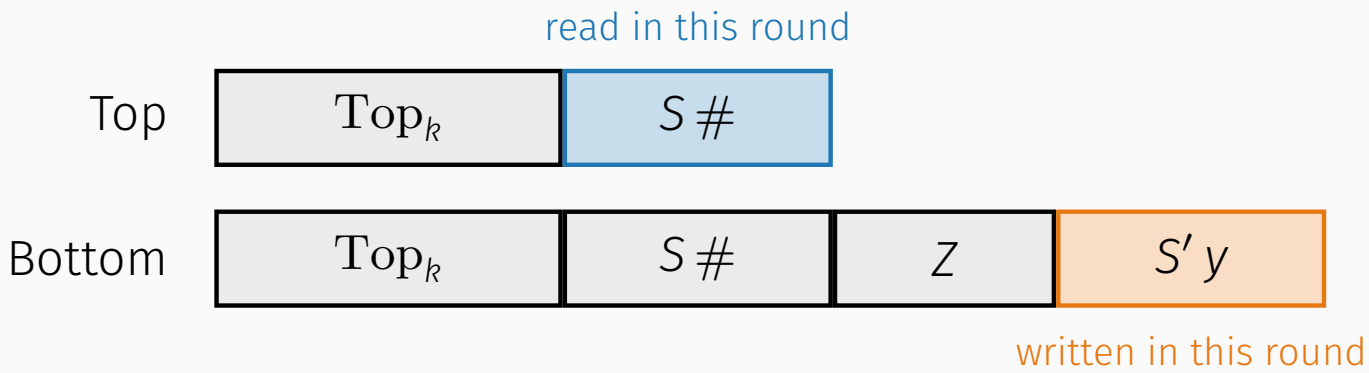
Facts. (1) Γ, Q, $\{\#\}$ are disjoint. (2) Every domino has nonempty top and bottom. (3) M is deterministic: $S \approx C_t$, $S \vdash S' \Rightarrow S' \approx C_{t+1}$.

Key Lemma: One Round

Lemma 3
Suppose $\mathrm{Bot}_k = \mathrm{Top}_k\, S\, \# \, Z$, *where* S *is a configuration string with state* $q \neq q_{accept}$. *Then* $q \neq q_{reject}$, *and there exist* $k' \in (k, m]$ *and* S' *with* $S \vdash S'$ *such that*

$$\mathrm{Top}_{k'} = \mathrm{Top}_k\, S\, \#, \qquad \mathrm{Bot}_{k'} = \mathrm{Bot}_k\, S'\, y,$$

where $y \in \{\#,\ \sqcup\#,\ \#\, q_0 w\, \#\}$.



Proof of the Key Lemma (1/2)

1. **The top must spell $S\#$ next.** $|\mathrm{Bot}_k| > |\mathrm{Top}_k|$ and $\mathrm{Top}_m = \mathrm{Bot}_m$, so $k < m$. Since $\mathrm{Top}_k\, S\#$ is a prefix of W, the tops of $D_{k+1}, D_{k+2}, \ldots$ spell $S\# \cdots$.
2. **The round ends exactly at $\#$.** Let $D_{k'}$ ($k' > k$) be the domino whose top covers the $\#$ after S. Tops containing $\#$: $\#$ (Parts 1, 5) and $q_{\text{accept}}\#\#$ (Part 7). Part 7 needs q_{accept} or $\#$ right before this $\#$; S ends with neither. So $D_{k'}$ has top $\#$, and $\mathrm{Top}_{k'} = \mathrm{Top}_k\, S\#$.

Proof of the Key Lemma (2/2)

3. **Only Parts 2, 3, 4 in between.** The tops of $D_{k+1}, \ldots, D_{k'-1}$ concatenate to S: no $\#$, no q_{accept}. This rules out Parts 1, 5, 6, 7.

4. **Exactly one transition domino** X (count state symbols); the rest are $\left[\frac{a}{a}\right]$. With $S = u \cdot \mathrm{top}(X) \cdot v$, $u, v \in \Gamma^*$:

   $$X = \left[\frac{qa}{br}\right] \Rightarrow S' = u\,br\,v; \qquad X = \left[\frac{cqa}{rcb}\right] \Rightarrow S' = u\,rcb\,v.$$

   Either way $S \vdash S'$; and X exists only if $q \notin \{q_{\text{accept}}, q_{\text{reject}}\}$.

5. **The last domino** $D_{k'} \in \left\{d_1, \left[\frac{\#}{\#}\right], \left[\frac{\#}{\sqcup\#}\right]\right\}$ appends y. $\square$

Finishing the Proof

Invariant $(*)$ at index k:

$$\mathrm{Bot}_k = \mathrm{Top}_k \; S_1 \# S_2 \# \cdots S_p \#,$$

where $p \geq 1$ and each $S_i \approx C_t$ for some t.

- **Base** $k = 1$: $\mathrm{Top}_1 = \#$, $\mathrm{Bot}_1 = \# \, q_0 w \, \#$, and $q_0 w = C_0$.
- **Step:** if $S_1 \approx C_t$ is not in q_{accept}, the Lemma gives $k' > k$ with $(*)$ for

  $S_2, \ldots, S_p$, S' (or $S'\sqcup$) (and possibly $q_0 w$), $\qquad S' \approx C_{t+1}$.

- **End:** indices increase but never exceed m, so the steps stop. They can only stop when $S_1 \approx C_t$ is in state q_{accept}.

So M reaches q_{accept} on w: M accepts w. $\square$